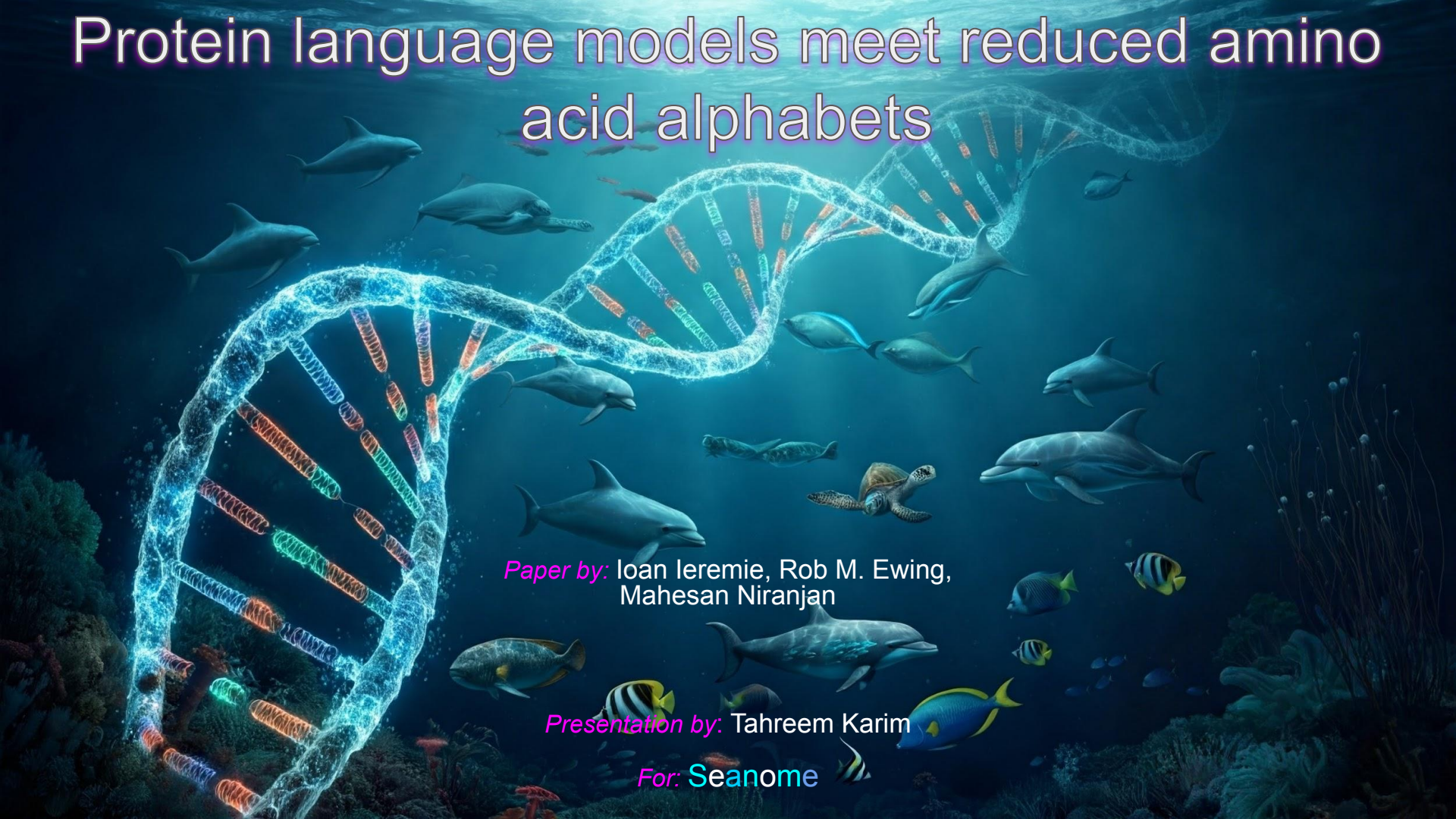
Protein language models meet reduced amino acid alphabets



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Review

Reduced alphabet - a simplified representation of amino acids in protein sequences. The 20 standard amino acids are grouped into a smaller number of clusters based on shared biochemical or structural properties. This *reduction aims to lower computational complexity while preserving critical information* for predictions and analyses.

Protein learning models (PLMs*)* - Language models are adapted for protein learning by treating protein sequences as a "language" and applying similar methods to learn meaningful representations of these sequences.

> *Each amino acid is treated as a "token"* - individual units of information, much like words in a sentence in natural language processing (NLP).

Methodology

Train Protein language model on **reduced alphabet structure** →

Translate amino acid sequence of interest using same reduced alphabet →

Input into trained Protein language model → Output structure prediction →

Assess accuracy →

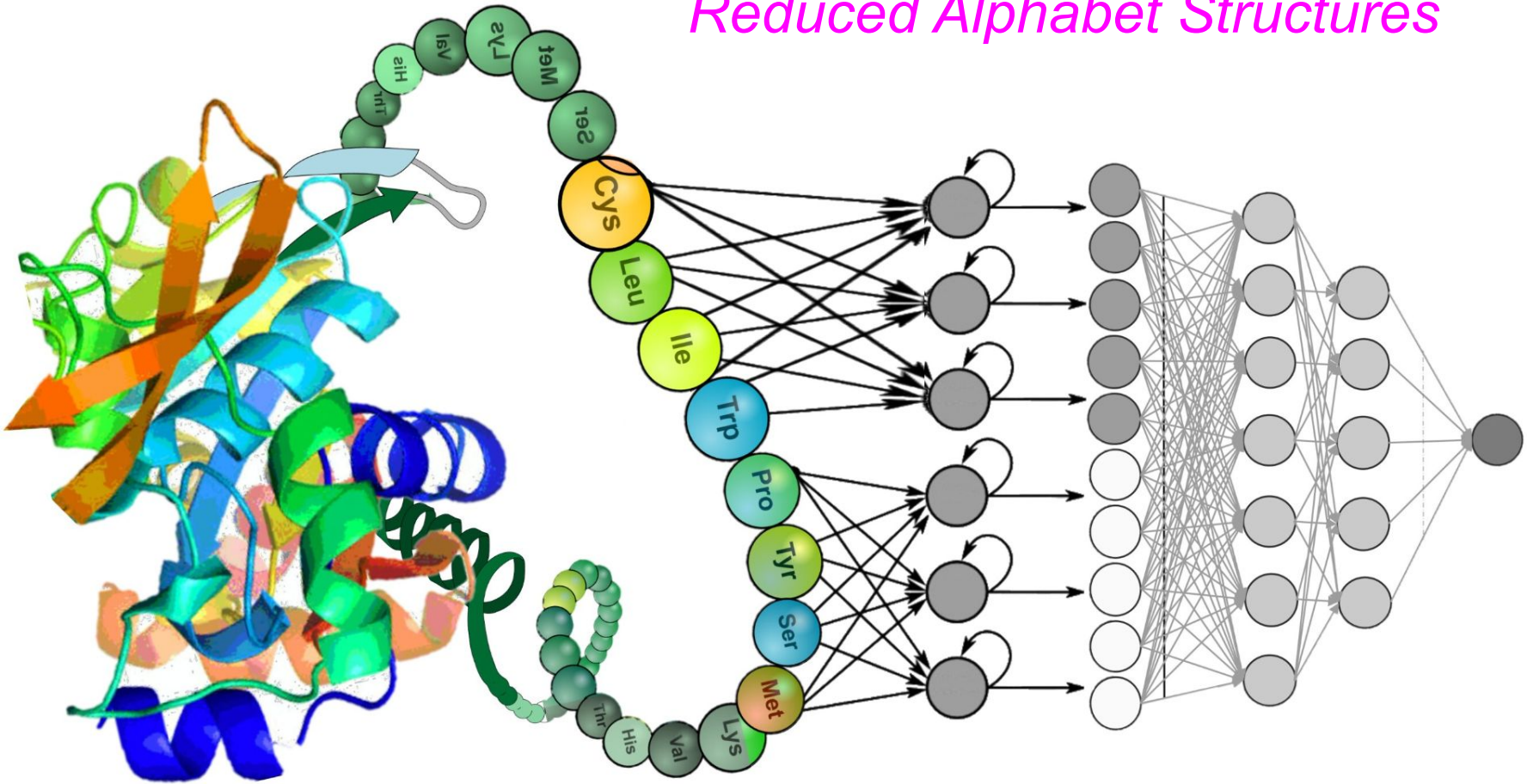
Compare to structure predicted by inputting **full AA sequence**

into PLM trained on **full alphabet** →

Compare accuracy of both

- This paper compared the *efficacy of various reduced alphabets when capturing evolutionary information and predicting protein structure via ESM Fold*.

Reduced Alphabet Structures



R.A.'s contd..

- Reduced alphabets will cluster amino acids based on physical and chemical properties like hydrophobicity, charge, polarity, size, and aromaticity.

- Alphabets used to train the protein learning model and respective clustering:
  - **WWMJ5** - hydrophobicity, polarity, and side-chain characteristics
  - **GBMR4** & **GBMR7** - substitution similarity under evolutionary constraints
  - **SDM12** - substitution matrices like *BLOSUM or PAM*
  - **HSDM 17** - hydrophobicity scales and substitution patterns
  - **MMSEQ 12** - amino acids frequencies and substitution probabilities observed in large protein sequence datasets.
  - **WASS14** - based on Wang and Wang's (2009) amino acid substitution matrix, which is grounded in evolutionary substitution patterns. It groups amino acids with similar biochemical properties (e.g., charge, hydrophobicity) and substitution tendencies observed in evolutionary processes.
  - **Uniprot18** - encodes the Pro-Glu and Leu-His amino acid pairs the same way.

Results

- **WASS14** and **MMSEQS12** outperforming full alphabet on specific AAV splits suggests that *alphabet reduction can act as a form of regularization by collapsing (biochemically) similar amino acids*. The model is forced to focus on mutations that cross cluster boundaries, which may be more functionally meaningful for AAV capsid fitness.

- WASS14's strong performance relative to alphabets of similar size is notable because its *design based on evolutionary substitution patterns*. **Hydrophobicity** appears better suited to capturing functionally relevant variation than purely frequency based schemes like MMSEQS12.

- “As PLM are trained to understand **sequence context**, proteins encoded with highly reduced alphabets do not provide enough diversity to allow the learning of **complex relations**.”

Results contd..

- The **error rate** in distinguishing mutations from the wild type increases as the alphabet reduces

- “Our empirical work shows that PLMs trained on the full alphabet and a large number of sequences capture **fine details** that are lost in alphabet reduction methods.”

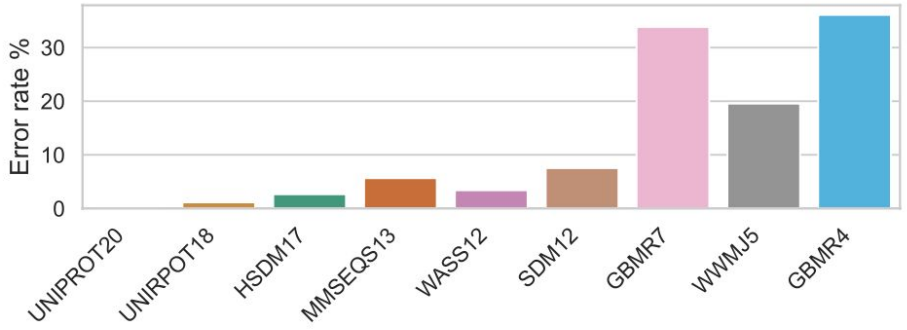


Figure 4. The error rate of the amino acid alphabets as the percentage of mutation pairs from the AAV dataset that cannot be distinguished. Alphabets with a few clusters are unable to capture the effect of mutations. Notice the low number of errors made by the WASS14 alphabet compared to alphabets of similar size due to its design based on hydrophobicity.

- Predictions made by **ESM Fold** - a protein structure prediction model that *does not rely on multiple sequence alignments* (**MSAs**).
    - A departure from traditional methods like Alpha Fold, which depend heavily on MSAs to capture evolutionary relationships between sequences.

- *LDDT-Cα* (Local Distance Difference Test for Alpha Carbons) a metric for evaluating the accuracy of predicted protein structures by comparing them to experimentally determined structures (e.g., from X-ray crystallography). Specifically, how well the *distances between alpha carbon (Cα) atoms* in the **predicted** structure match those in the **reference** structure.

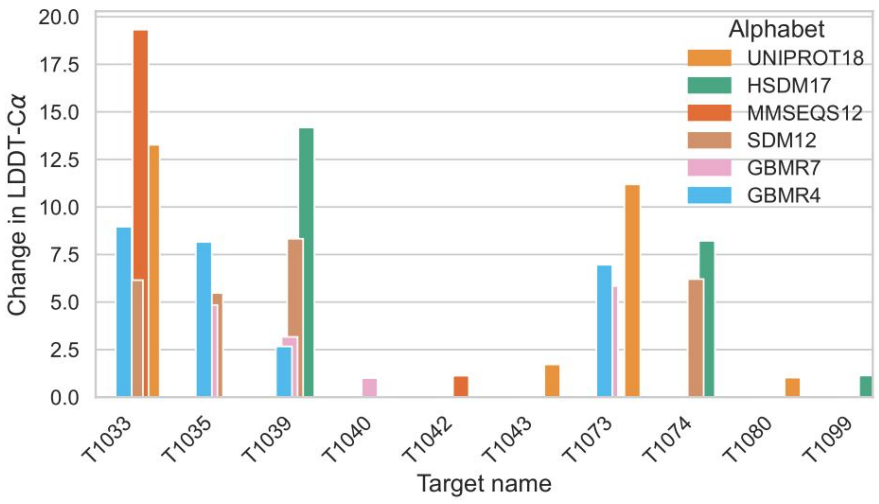


Figure 7. Ten targets from CASP14 for which the alphabet reduction methods improve the overall LDDT-Cα scores by a margin of at least 1%. The target T1039, which belongs to the FM (free modelling) category, is better predicted by ESMfold when using the HSDM17 alphabet, with an increase of LDDT-Cα of 14%. Similarly, for the T1033 target, the difference in LDDT-Cα score to the original sequence is 19%. For the target T1035, small alphabets show improvements in LDDT-Cα over 5%.

CASP14 - essentially *a collection of challenging protein structure prediction tasks. It serves as a global benchmarking event* to test and compare the accuracy of computational methods in predicting the 3D structures of proteins under realistic and blind conditions.

Discussion

- "*Surprisingly, alphabets with >10 clusters perform better than the full alphabet on certain splits*. Namely, **WASS14** on the '2/7-versus-many AAV' splits, and **MMSEQS12** on the '7-versus-many AAV' split.
    - This suggests that while the full alphabet has the ability to distinguish between all mutation pairs that appear in the dataset, in certain cases ignoring those mutations and focusing on **out-of-clusters mutations** improves performance."

- This finding shouldn't be over generalized, it is dataset specific. *The full alphabet still dominates on most tasks*, and the AAV dataset's mutation distribution likely explains why these two alphabets excel on those splits.

- Would these results hold on other viral capsid datasets, or is this specific to AAV's constrained sequence?

Topics the paper also covers

$$\text{Perplexity}(x) = \exp\{-\log p(x_{i \in M} | x_{j \notin M})\}$$

However, perplexity seems to be a good proxy measure of 'understanding' proteins and their structures, with models that perform better on the language task showing an increased ability to predict protein structures (Lin *et al.* 2022).

$$\text{PPPL}(x) = \exp\left\{-\frac{1}{L}\log p(x_i | x_{\neg i})\right\}$$

- Training step, validating perplexity
- *PLM fold prediction accuracy* - as you'd expect, the less you reduce the alphabet, the better the PLM accuracy. **UNIPROT 20** was the best.
- Multi Seq Analysis of modified sequences

Future directions

- Since our primary focus is the protein function rather than the **evolutionary origins** of genes, using a reduced alphabet can help minimize computational power requirements while maintaining accurate prediction models.

- Eventually, ML models may be able to *predict protein mechanisms from sequences involving several proteins* by training on metagenomics & protein interactions.

- There is huge potential for **Synthetic biology**. ML models may eventually be able to assess the viability of novel organisms before attempting to synthesize them in the lab.

  - All this could be part of AI tech that helps build/troubleshoot synthetic organisms.

Sources

Cover slide photo: generated by gemini “Ocean creatures swimming and DNA”

Reduced alphabet picture:
https://phys.org/news/2022-04-deep-decode-functional-properties-proteins.html